SALES PERFORMANCE ANALYSIS AND REVENUE OPTIMIZATION USING DATA ANALYTICS

A PROJECT REPORT

PRESENTED TO THE AICIC CONCEPTS LIMITED, UNIZIK AWKA, ANAMBRA STATE

IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE MATRIX COHORT 2026

BY

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NOVEMBER, 2026

DECLARATION

I hereby declare that this project titled “Sales Performance Analysis and Revenue Optimization Using Data Analytics” is my work and has been prepared for academic purposes. All sources consulted in the course of the project have been acknowledged.

Name: Apakama Ugochukwu Chimamkpam

Department Of Data Analysis

Signature: _____ Date: _____

CERTIFICATION

This is to certify that this project titled “Sales Performance Analysis and Revenue Optimization Using Data Analytics” was carried out by Apakama Ugochukwu Chimamkpam, Department of Data Analysis at AICIC Concepts LTD.

Project Supervisor: _____

Signature: _____ Date: _____

DEDICATION

This project is dedicated to God Almighty for His grace, strength and guidance throughout the period of this work, and to my family for their support and encouragement.

ACKNOWLEDGEMENT

I sincerely thank God Almighty for the wisdom, strength and ability to complete this project. My appreciation goes to my project supervisor for the guidance and corrections provided during the course of this work. I also appreciate the management and staff of AICIC Concepts LTD, for the knowledge and support received during my training. Finally, I thank my family, friends and everyone who contributed directly or indirectly to the successful completion of this project.

ABSTRACT

This project examined sales performance and opportunities for revenue optimization using data analytics. A Nigerian E-Commerce Dataset obtained from Kaggle was used for the analysis. The dataset was cleaned and prepared with Microsoft Excel Power Query, analysed using Microsoft SQL Server, and presented through interactive dashboards in Microsoft Power BI. The analysis focused on order status, revenue, quantity sold, monthly performance, products, vendors, branches and customer regions. The cleaned dataset contained 3,928 records, while the delivered records generated a total revenue of ₦1,227,420,031.43 from 1,055 distinct orders and 142,867 units. May 2021 recorded the highest monthly revenue, Milo Sachet 20g was the leading product by revenue, and Renuzi Ventures was the highest-performing vendor. Lagos accounted for the largest share of branch and regional revenue. The findings demonstrate how spreadsheet, database and business intelligence tools can support sales monitoring and evidence-based business decisions.

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Businesses generate large amounts of data through sales transactions, customer orders, products and other daily activities. When properly collected and analysed, this data can provide useful information about business performance and help management make informed decisions.

Sales performance analysis involves examining sales records to understand revenue, quantity sold, order volume, product performance and market trends. Data analytics tools make it easier to organise large datasets, identify patterns and present findings in a clear format.

This project uses a Nigerian E-Commerce Dataset to examine sales performance between February and May 2021. Microsoft Excel was used for data cleaning and preparation, Microsoft SQL Server for structured querying and analysis, and Microsoft Power BI for dashboard presentation.

1.2 Statement of the Problem

Sales records may contain missing values, inconsistent entries and large volumes of information that are difficult to interpret manually. Without proper analysis, a business may find it difficult to identify its best-performing products, vendors, branches and sales periods. This can affect planning, stock management and revenue improvement. The project therefore applies data analytics tools to organise sales information and produce useful findings.

1.3 Aim and Objectives

The aim of this project is to analyse sales performance and identify opportunities for revenue optimization using data analytics.

The specific objectives are to:

- clean and prepare the Nigerian E-Commerce Dataset;
- analyse sales revenue, quantity sold and order volume;
- identify top-performing and low-performing products and vendors;
- examine monthly, branch and regional sales performance;
- develop Power BI dashboards to present the findings; and
- provide recommendations based on the results.

1.4 Research Questions

1. What was the overall sales performance recorded in the dataset?
2. Which months recorded the highest sales revenue and order volume?
3. Which products and vendors generated the highest revenue?
4. How did sales performance differ across branches and customer regions?
5. How can the findings support better business decisions?

1.5 Significance of the Study

This project demonstrates the practical use of Excel, SQL Server and Power BI in analysing business data. It can help business owners and decision-makers understand sales trends, monitor product and vendor performance, identify areas requiring attention and make better-informed decisions. It also provides practical experience in data cleaning, database querying and data visualization.

1.6 Scope of the Study

The study focuses on the Nigerian E-Commerce Dataset covering the period from 1 February 2021 to 31 May 2021. The analysis considers order status, delivered revenue, quantity sold, order volume, average order value, product performance, vendor performance, branch location, customer region and local area. The tools used are Microsoft Excel, Microsoft SQL Server and Microsoft Power BI.

1.7 Limitations of the Study

The project is limited by the period and fields available in the dataset. Some geographical records were labelled as Unknown, and the dataset covers only four months in 2021. The findings should therefore be interpreted within the limits of the available data and should not be treated as a description of current or year-round sales.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter explains the major concepts related to data analytics, sales performance analysis and revenue optimization. It also describes the software tools used in carrying out the project.

2.2 Concept of Data Analytics

Data analytics is the process of examining data to discover patterns, trends and useful information. It involves activities such as collecting, cleaning, transforming, analysing and presenting data. In business, analytics can support planning, performance monitoring and decision-making.

2.3 Sales Performance Analysis

Sales performance analysis is the examination of sales information to determine how well a business, product, vendor or location is performing. Common measures include total revenue, quantity sold, number of orders, average order value and sales growth. Comparing these measures over time can reveal changes in demand and business performance.

2.4 Revenue Optimization

Revenue optimization refers to the use of business information and strategies to improve the income generated from sales. It may involve promoting products with strong demand, improving product availability, reviewing vendor performance, reaching new markets and encouraging customers to purchase complementary products. Data analysis helps identify where such actions may be useful.

2.5 Tools Used in the Project

2.5.1 Microsoft Excel

Microsoft Excel is a spreadsheet application used to organise, calculate, filter and analyse data. Power Query, a feature in Excel, supports the import, cleaning and transformation of data. In this project, Excel Power Query was used to prepare the dataset before analysis.

2.5.2 Microsoft SQL Server

Microsoft SQL Server is a relational database management system used to store and manage structured data. SQL commands can be used to retrieve, filter, group and summarise records. In this project, SQL Server was used to analyse sales revenue, orders, products, vendors and geographical performance.

2.5.3 Microsoft Power BI

Microsoft Power BI is a business intelligence tool used to transform data into reports and interactive visualizations. It supports charts, tables, cards and dashboards that help users understand key performance indicators. Power BI was used in this project to present the results of the sales analysis.

2.6 Importance of Data Analytics in Business

Data analytics helps businesses make decisions based on recorded evidence rather than assumptions. It can reveal sales patterns, highlight products and locations that contribute strongly to revenue, support inventory planning and identify possible data-quality problems. Clear visual reports also make findings easier to communicate.

2.7 Chapter Summary

This chapter reviewed data analytics, sales performance analysis and revenue optimization. It also discussed Excel, SQL Server and Power BI, which were combined in this project to prepare, analyse and present sales data.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter describes the methods and tools used to complete the project. The work involved obtaining the dataset, cleaning and preparing the records, analysing the data with SQL Server and presenting the results with Power BI.

3.2 Research Design

The project adopted a quantitative, descriptive data-analysis approach. Numerical sales records were examined to summarise business performance and identify patterns across time, products, vendors and locations.

3.3 Data Source and Description

The dataset used for this project was the Nigerian E-Commerce Dataset obtained from Kaggle. The original data was prepared for analysis and saved as SalesData_Cleaned.csv. The imported SQL Server table contained 3,928 records and 16 columns, including Order_ID, Branch_Location, Branch_Name, Business_Name, Item_ID, Item_Name, Item_Price, Item_Status, Quantity, Total_Price, Order_Date, Order_Region and Order_Local_Area.

The records covered the period from 1 February 2021 to 31 May 2021. The analysis of realized sales focused on records with an Item_Status of Delivered.

3.4 Data Cleaning and Preparation

Microsoft Excel Power Query was used to inspect and prepare the dataset. The cleaning process involved reviewing columns, checking missing or inconsistent values, applying suitable data types and preparing the cleaned file for import into SQL Server. The cleaned dataset was saved as SalesData_Cleaned.csv.

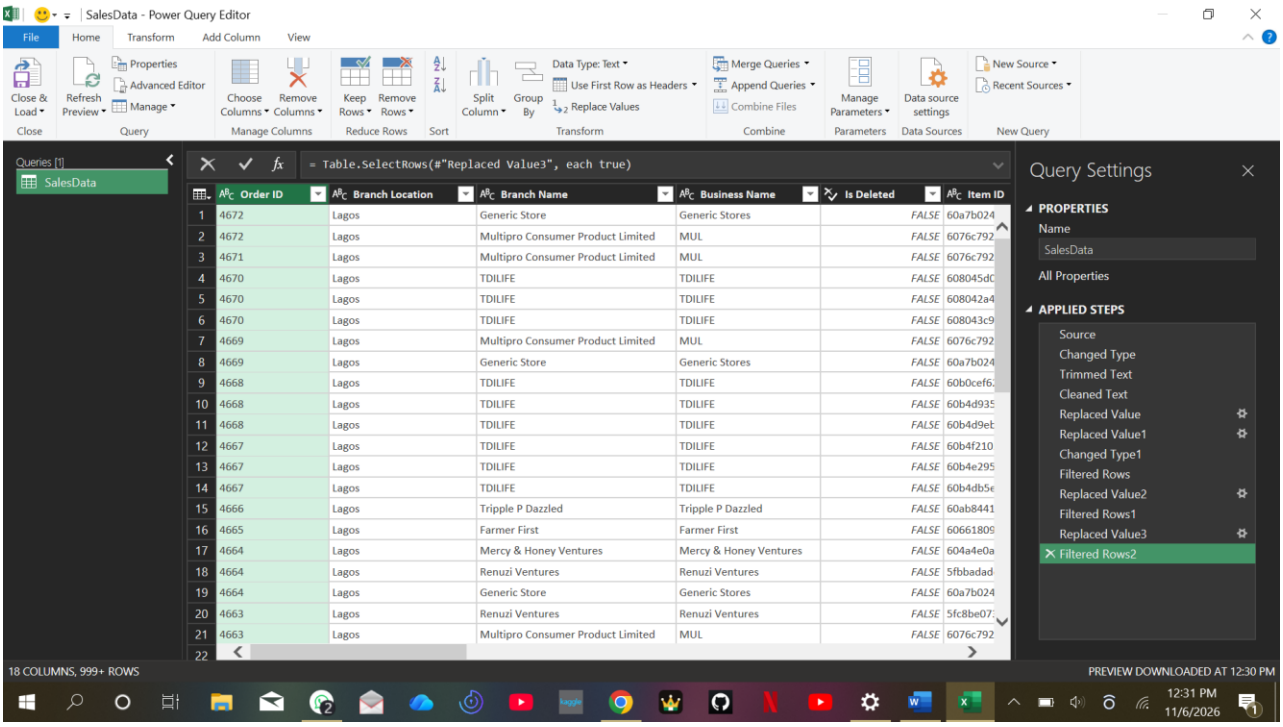


Figure 3.1: Data Cleaning and Transformation Using Microsoft Excel Power Query.

3.5 SQL Server Analysis

The cleaned CSV file was imported into a database named SalesPerformanceDB in Microsoft SQL Server. The table used for analysis was dbo.SalesData_Cleaned. SQL queries were written to examine order status, total delivered revenue, quantity sold, distinct orders, average order value, monthly sales, product performance, vendor performance, branch performance, regional performance and data quality.

The analysis used filtering and aggregation operations such as WHERE, GROUP BY, ORDER BY, COUNT, SUM and AVG. Delivered records were used for the main sales performance calculations so that cancelled and other unsuccessful statuses were not counted as completed sales.

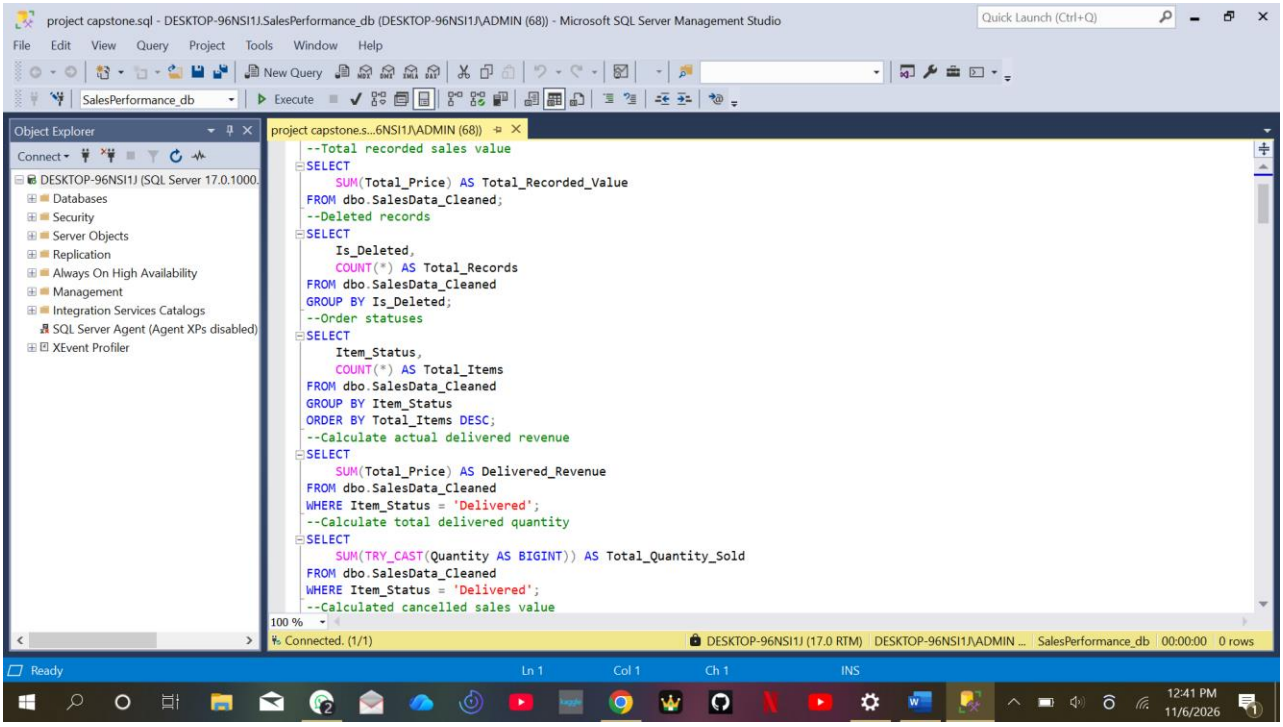


Figure 3.2: Sales Data Analysis Using Microsoft SQL Server Management Studio.

3.6 Power BI Visualization

Microsoft Power BI Desktop was connected to the SalesPerformanceDB database and the dbo.SalesData_Cleaned table. Measures were created to calculate total revenue, total quantity sold, total orders and average order value for delivered records. Charts and KPI cards were then arranged across two report pages: Sales Performance Overview and Detailed Sales Performance Analysis.

3.7 Chapter Summary

This chapter explained the data source, preparation process and analytical tools used in the project. Excel Power Query was used for cleaning, SQL Server for querying and analysis, and Power BI for visual presentation.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results obtained from analysing the Nigerian E-Commerce Dataset. The analysis focuses on delivered sales records and examines order status, revenue, monthly performance, products, vendors, branches and customer regions.

4.2 Data Presentation

A total of 3,928 records were imported into SQL Server. The distribution of records by order status is presented below.

Order Status	**Number of Records**
Delivered	1,655
Cancelled	2,041
Invalid Order	227
Failed Delivery	4
Packed	1
Total	3,928

4.3 Analysis of Sales Performance

4.3.1 Overall Sales Performance

Performance Measure	**Result**
Total delivered revenue	₦1,227,420,031.43
Total quantity sold	142,867
Distinct delivered orders	1,055
Average order value	₦1,163,431.31
Unique products sold	270

The delivered records generated over ₦1.2 billion in revenue from 1,055 distinct orders. The results provide an overall view of the sales performance within the period covered by the dataset.

4.3.2 Monthly Sales Performance

Month (2021)	Quantity Sold	Orders	Revenue
February	10,453	32	₦112,892,680.00
March	35,016	107	₦232,444,352.00
April	52,272	288	₦424,428,253.43
May	45,126	628	₦457,654,746.00

Revenue increased from February to May, with May recording the highest revenue and number of orders. April recorded the highest quantity sold. February had the lowest revenue and order volume.

4.3.3 Product Performance Analysis

Product	Quantity	Revenue
Milo Sachet 20g	20,056	₦183,035,259.00
Knorr Chicken 17x50x8g	13,464	₦127,264,413.00
Milo 20g	8,891	₦80,281,650.00
Maize	3,890	₦74,749,000.00
Close Up Ever Fresh Red Hot 140g	5,086	₦70,583,650.00

Milo Sachet 20g recorded the highest revenue among the products analysed, followed by Knorr Chicken 17x50x8g and Milo 20g.

4.3.4 Vendor Performance Analysis

Vendor	Delivered Revenue
Renuzi Ventures	₦387,313,524.00
Tripple P Dazzled	₦374,859,929.50
Farmer First	₦104,334,920.00
Deeski Store Limited	₦80,089,178.00
Mercy & Honey Ventures	₦75,869,465.00
Product Plug	₦19,260.00

Renuzi Ventures recorded the highest vendor revenue, while Product Plug recorded the lowest revenue among the vendors listed in the analysis.

4.3.5 Branch Performance Analysis

Branch Location	Orders	Quantity	Revenue
Lagos	1,048	136,355	₦1,196,409,216.43
Oyo	18	6,512	₦31,010,815.00

The Lagos branch generated the largest share of delivered revenue and accounted for most of the recorded orders and quantity sold.

4.3.6 Regional Sales Performance

Customer Region	Quantity	Revenue
Lagos	130,955	₦1,119,709,531.18
Unknown	4,298	₦47,338,162.00
Ogun	4,147	₦35,638,523.25
Oyo	3,467	₦24,733,815.00

Lagos was the leading customer region by revenue. The presence of Unknown values indicates that some geographical information was not available in the dataset.

4.3.7 Local Area Performance

Local Area	Revenue
Oshodi-Isolo	₦349,298,700.55
Unknown	₦285,510,124.00
Alimosho	₦209,339,520.93
Ado-Odo/Ota	₦84,794,118.50
Mushin	₦68,212,926.00

Oshodi-Isolo recorded the highest revenue among the listed local areas, followed by Unknown and Alimosho.

4.3.8 Average Order Value Analysis

Month	Number of Orders	Average Order Value
February	32	₦3,527,896.25
March	107	₦2,172,377.12

April	288	₦1,473,709.21
May	628	₦728,749.60

Average order value declined each month while the number of orders increased. This indicates that the average value per order became lower as order volume grew.

4.4 Data Quality Assessment

The data-quality checks showed that all 3,928 records had Is_Deleted set to 0. Among delivered records, no invalid quantity values, missing item names or invalid, missing, zero or negative total-price values were found. However, 56 repeated Order_Item_Number identifiers affected 112 delivered records. These were treated as records requiring further investigation rather than confirmed duplicates.

4.5 Power BI Dashboard Presentation

Two report pages were created in Power BI to present the results in a clear visual format.

4.5.1 Sales Performance Overview

The first page presents the main performance indicators, including total revenue, total quantity sold, total orders and average order value, together with monthly revenue and monthly order volume.

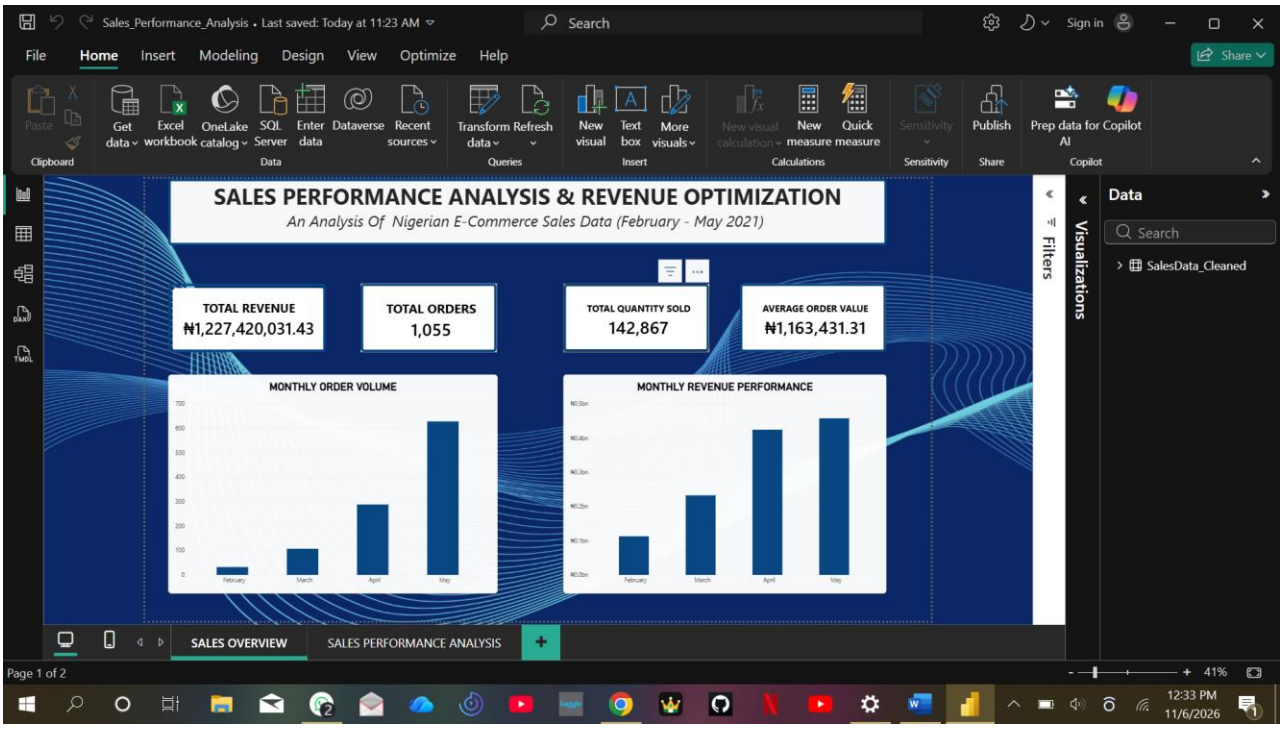


Figure 4.1: Sales Performance Overview Dashboard.

4.5.2 Detailed Sales Performance Analysis

The second page presents detailed charts covering top products, top vendors, branch revenue, customer-region revenue, local-area revenue and monthly average order value.

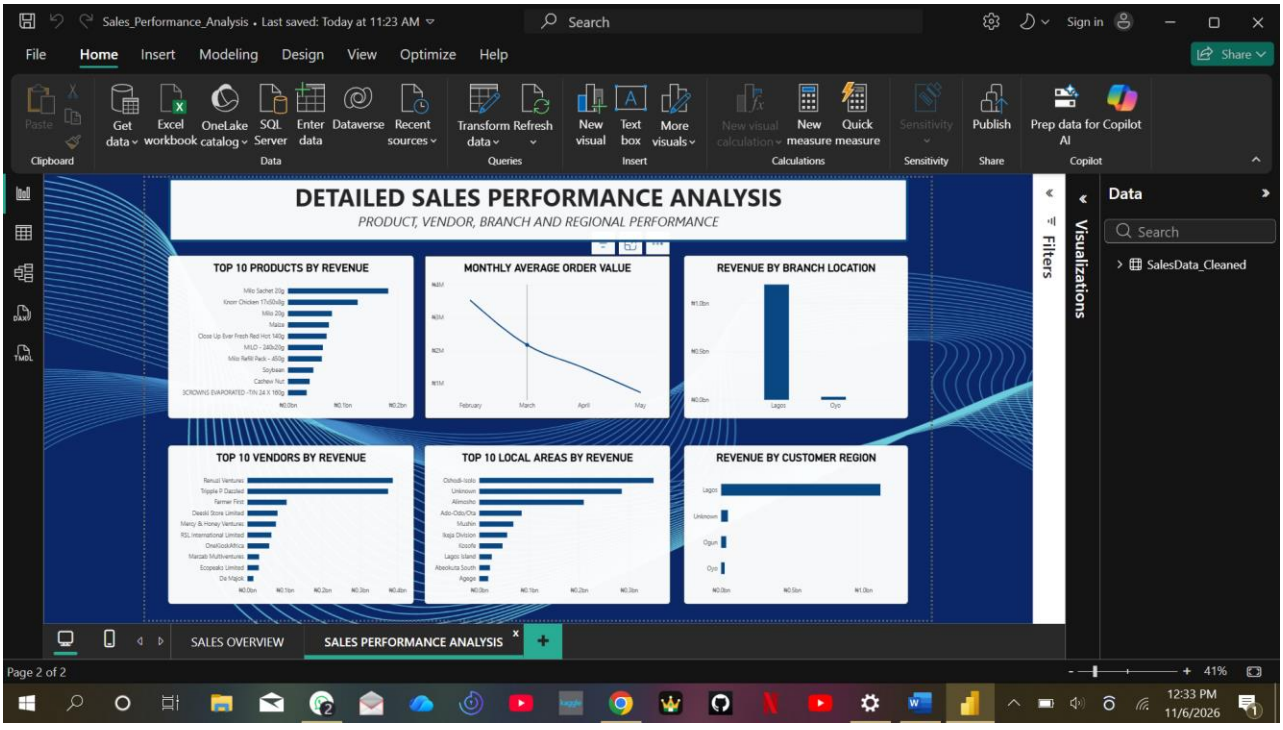


Figure 4.2: Detailed Sales Performance Analysis Dashboard.

4.6 Discussion of Findings

The analysis showed that sales revenue increased throughout the four-month period, although average order value declined as order volume increased. A small number of products, vendors and locations contributed a large share of the recorded revenue. Lagos was the dominant branch location and customer region, while Milo Sachet 20g was the leading product by revenue. The data-quality checks also identified incomplete geographical information and repeated order-item identifiers that should be reviewed.

4.7 Chapter Summary

This chapter presented the results of the sales analysis and the Power BI dashboards. The findings provide information about overall sales, monthly trends, product and vendor performance, geographical performance and data quality.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary

This project focused on analysing sales performance and identifying opportunities for revenue optimization using data analytics. The Nigerian E-Commerce Dataset was cleaned and prepared using Microsoft Excel Power Query, analysed using Microsoft SQL Server and presented using Microsoft Power BI.

The analysis recorded delivered revenue of ₦1,227,420,031.43, with 142,867 quantities sold and 1,055 distinct orders. May 2021 recorded the highest monthly revenue and order volume, while April recorded the highest quantity sold. Milo Sachet 20g was the highest-performing product by revenue, Renuzi Ventures was the highest-performing vendor, and Lagos accounted for the largest share of branch and regional revenue.

5.2 Conclusion

The project demonstrated how data analytics tools can be used to clean, examine and present sales records. The use of Excel, SQL Server and Power BI made it possible to identify sales trends, top-performing products and vendors, and geographical patterns. The findings can support better sales monitoring, inventory planning and business decision-making.

5.3 Recommendations

6. Businesses should maintain sufficient stock of high-performing products to meet demand.
7. Vendor performance should be reviewed regularly to identify reliable suppliers and areas requiring improvement.
8. Businesses should explore ways to improve sales in branches and regions with lower performance.
9. Promotions, product bundles and cross-selling can be considered to improve average order value.
10. Data collection processes should be improved to reduce missing customer-location and order information.
11. Repeated order-item identifiers should be investigated before records are changed or deleted.

5.4 Suggestions for Further Study

Future studies may use datasets covering more years to examine long-term sales trends. Further analysis could also investigate customer buying behaviour, seasonal demand and other factors that influence sales performance.

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